Task 1: QA & Debug Report

Prepared by Shashwat Gupta | Automation & QA Developer take-home assessment

1. Scope and environment

App tested: Conduit (RealWorld Angular frontend) at https://demo.realworld.show, the replacement demo listed in the README of github.com/gothinkster/realworld.

Note: the original demo, https://demo.realworld.io, was offline during testing and returned a 404 'NoSuchBucket' error (screenshot 01).

Method: Google Chrome on Windows. Manual walk-through of sign-up, login, create article, validation, delete and logout, plus a Lighthouse audit (Accessibility, SEO, Performance, Best Practices) run from Chrome DevTools.

Severity scale: Critical = data loss, security hole or app unusable. High = a main flow is broken. Medium = wrong behaviour with a workaround. Low = cosmetic or minor UX.

2. Summary

11 issues found: 0 Critical, 0 High, 4 Medium, 7 Low. No Critical or High issues were found in the flows tested. The most important finding for real users is #11 (deleting an article has no confirmation). Accessibility is the weakest area (Lighthouse Accessibility score 67). Items marked 'suspected cause' are hypotheses based on observed behaviour; I did not have access to the source code.

3. Bug table

#	Title / Summary	Steps to Reproduce	Expected vs Actual	Severity	Suspected Cause
1	Low text contrast on muted text (article descriptions, 'Read more...', tags, nav links, Edit Article button)	1. Open https://demo.realworld.show 2. F12 > Lighthouse > tick Accessibility > Analyze page load 3. Open 'Background and foreground colors do not have a sufficient contrast ratio'	**Expected:** text meets WCAG AA contrast (4.5:1). **Actual:** light grey text on white fails the audit and is hard to read.	Medium	Light grey 'muted' colour used on a white background without a contrast check.
2	Images have no alt attribute	Same Lighthouse run: Accessibility > 'Image elements do not have [alt] attributes' (also reported under SEO).	**Expected:** every image has alt text (or empty alt if decorative). **Actual:** alt missing, so screen readers cannot describe the images.	Medium	Image tags (probably user avatars) rendered without an alt binding.
3	Links without a discernible name	Same Lighthouse run: Accessibility > 'Links do not have a discernible name'.	**Expected:** every link has readable text or an aria-label. **Actual:** some links have no text a screen reader can announce.	Medium	Icon-only or image-only links with no text or aria-label.
4	<html> element has no lang attribute	Same Lighthouse run: Accessibility > Internationalization and localization.	**Expected:** <html lang="en">. **Actual:** attribute missing.	Low	Missing from the base index.html.
5	No meta description	Lighthouse > SEO > 'Document does not have a meta description'.	**Expected:** a short description for search results. **Actual:** none.	Low	Tag not added to the page head.

#	Title / Summary	Steps to Reproduce	Expected vs Actual	Severity	Suspected Cause
6	robots.txt is invalid (14 errors)	Lighthouse > SEO > 'robots.txt is not valid'. Also open https://demo.realworld.show/robots.txt in the browser.	**Expected:** a valid plain-text robots.txt. **Actual:** Lighthouse reports 14 errors.	Low	Server may return the app's HTML page for /robots.txt (single-page-app fallback). Not yet verified.
7	Layout shift on load (CLS 0.113) and render-blocking requests	Lighthouse > Performance: 'Cumulative Layout Shift', 'Layout shift culprits', 'Render-blocking requests' (est. 450 ms).	**Expected:** CLS under 0.1, content does not jump while loading. **Actual:** CLS 0.113 (flagged orange).	Low	Images without explicit width/height and late-loading fonts or content. Note: the overall Performance score varied between runs (88 and 74), so only the specific findings are reported.
8	Thick focus outline appears on mouse click and overlaps the icon in header links	1. Log in 2. Click 'New Article' in the header with the mouse	**Expected:** no ring on mouse click (ring only for keyboard Tab); if shown, it should have spacing around icon and text. **Actual:** a thick black box appears on click, touches the text, and the pencil icon overlaps its left edge.	Low	Custom outline probably applied with :focus instead of :focus-visible, with no padding or outline-offset.
9	Validation errors not linked to the fields they refer to	1. Log in 2. Click New Article 3. Leave everything empty and click Publish Article	**Expected:** invalid fields highlighted; messages use the same names as the form labels. **Actual:** a list at the top says 'description can't be blank', but no field is highlighted and the form has no field named 'description' (it says 'What's this article about?').	Low	Errors come from the API response and are shown as a plain list with the raw API field name.
10	Confusing error and no input limit for long article titles	1. Log in > New Article 2. Paste a title of 150+ characters; put 'test' in About and Body 3. Click Publish Article	**Expected:** the field stops at 100 characters or shows a counter, and the error says 'Title must be under 100 characters' next to the title. **Actual:** all input is accepted; after Publish the message reads 'body title is a string of less than 100 chars'.	Low	API validation message shown as-is (the word 'body' likely comes from a generic error key); no client-side maxlength.
11	Article is deleted immediately with no confirmation	1. Log in 2. Open one of your own articles 3. Click 'Delete Article'	**Expected:** a confirmation dialog ('Are you sure? This cannot be undone') or an undo option. **Actual:** the article is deleted instantly, with no popup.	Medium	The Delete button probably calls the delete API directly in its click handler, with no confirm step.

4. Root-cause analysis (Issue #1: low text contrast)

The muted text across the site (article descriptions, 'Read more...', tags, navigation links and the Edit Article button) is shown in a very light grey on a white background. Lighthouse flags this because the colour pair falls below the WCAG AA minimum contrast ratio of 4.5:1 for normal-size text. This most likely happens because the stylesheet uses one light 'muted' text colour for all secondary content, chosen for looks and never checked against the background. The impact is real: people with low vision, users on phones in bright sunlight, and anyone with a poor screen can barely read it. It also hurts ordinary users, because secondary text such as 'Read more...' is a key navigation cue. To fix it, I would find the CSS rule or variable that sets the muted colour and darken it. For example, #767676 on white is about 4.5:1, which is the minimum, and a darker grey gives more margin. Because one variable most likely controls it, a single change should fix every affected element. After the fix I would re-run Lighthouse to confirm the contrast audit passes, and add an automated accessibility check (for example axe in CI) so the problem does not come back.

5. What worked correctly

- Sign-up and creating a normal article worked; the article appeared under my username.
- Submitting an empty article form showed an error for each required field (title, description, body).
- Logout worked, and pressing the browser Back button afterwards led to the login page, not to a logged-in page.

6. Evidence (screenshots in the repository /screenshots folder)

- 01-demo-site-down.png: demo.realworld.io 404 NoSuchBucket
- 02-lighthouse-overview.png, 03-accessibility.png, 04-seo.png, 05-performance.png: Lighthouse results (issues 1-7)
- 06-focus-outline.png: focus outline on 'New Article' (issue 8)
- 07-validation-errors.png: empty-form errors (issue 9)
- 08-long-title-error.png: long-title error (issue 10)

7. Limitations

Testing was manual and limited to the main user flows on one browser. I did not test edit, comments, favourites, follow, mobile layouts or security headers in depth, and I did not inspect the source code, so all 'suspected causes' are hypotheses to be confirmed.